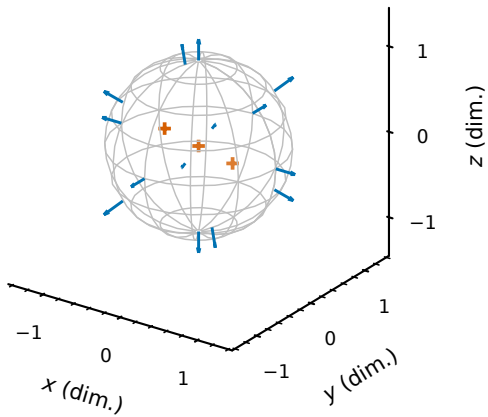
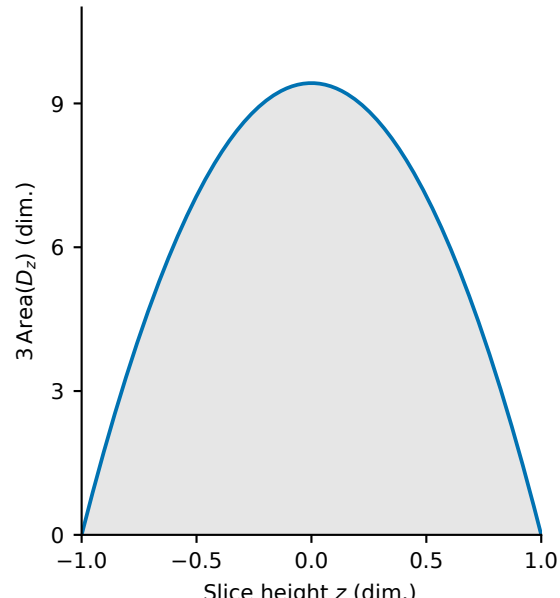
A



$F = r$: $\operatorname{div} F = 3$, $\Phi = 4\pi$

B



$\int_{-1}^{1} 3\pi(1 - z^2)\, dz = 4\pi$